Accessible Training for the Dyslexic Workforce: An Open-Source AI Blueprint

Michael King and Kevin Rose

Purdue University, West Lafayette, IN, USA

Copyright © 2026 Michael King & Kevin Rose

Abstract

Dyslexia is a neurodevelopmental learning disability that impacts how an individual reads and thinks (Panjwani-Charania & Zhai, 2024) and is characterized by marked difficulties in processing phonological information (Kizilaslan & Tunagür, 2021). Dyslexic individuals often experience slower reading speed, increased spelling mistakes, and increased difficulty with complex or technical documents (Schulte-Körne, 2010). These individuals are often perceived as "different, inferior, and stupid" (Ingesson, 2007, p. 583) despite comparable training and preparation. Prevalence estimates in English-speaking countries range between 5% and 17.5% (Kast et al., 2007), so organizations almost certainly employ dyslexic workers. However, many employees are hesitant to disclose their diagnosis to management due to the perceived stigma associated with the condition (Wissell et al., 2022).

Dyslexic individuals have notable working-memory limitations and frequently experience cognitive overload with solely text-based materials, depleting capacity otherwise used to process information (Kizilaslan & Tunagür, 2021). Consequently, dyslexic learners can comprehend and recall information better through the use of simplified language, shorter sentences in training materials (Kizilaslan & Tunagür, 2021), and multimedia presentation (Knoop-van Campen et al., 2018). For example, video that incorporates visual aids such as diagrams and images along with relevant audio helps people with dyslexia retain information better than text-based presentations (Le Cunff et al., 2025). These benefits align with Cognitive Load Theory (Sweller, 2011), multimedia learning theory (Mayer, 2020), and Universal Design for Learning (CAST, 2024), benefiting all learners without requiring disclosure. Current evidence is largely from studies of children, a gap this study addresses by investigating working adults.

This proposed study explores the feasibility of utilizing an open-source large language model (LLM) alongside a video diffusion model to bridge an accessibility gap in corporate instructional design. A system that condenses material and delivers it as multimedia could help dyslexic employees comprehend and retain training more efficiently than text or classroom formats (Kizilaslan & Tunagür, 2021; Knoop-van Campen et al., 2018).

This research outlines an innovative HRD workflow for the AI-assisted creation of training material. Gemma, an open-source LLM (Gemma Team, 2025), and LTX-2, an open-source text-to-video diffusion model (HaCohen et al., 2026), could be used in this pipeline. Relevant text is first extracted from the source PowerPoint slides; the LLM then adapts it for dyslexic learners while crafting specific prompts to create short training scenes in LTX-2. Finally, the videos are stitched into a professional training course custom-built for dyslexic audiences. All generated modules will undergo expert human review before learner use.

This study comprises a learner evaluation (RQ1–RQ2) and a pipeline evaluation (RQ3–RQ4):

- RQ1: To what extent does comprehension of training materials differ for dyslexic adults between AI-generated audiovisual and traditional text-based training?
- RQ2: To what extent does learner confidence in understanding differ between AI-generated audiovisual and traditional text-based training?
- RQ3: How accurately does the pipeline reproduce source content, quantified by error count, omissions, and hallucinations per module?
- RQ4: How does the pipeline impact production efficiency (hours and iterations per module) versus a conventional training workflow?

Research questions RQ1 and RQ2 are addressed using a within-subjects design (Charness et al., 2012), with each participant experiencing both formats in counterbalanced order. Target sample size will be determined through an a priori power analysis for the within-subjects comparison (80% power, medium effect). The study evaluates the complete intervention package of simplified language, audiovisual delivery, and restructured presentation rather than isolated components. Self-identified dyslexic adults will first complete the Adult Dyslexia Checklist (Smythe & Everatt, 2001) as a screening measure, with eligibility in the study requiring a score of 45 or higher on the checklist. Participants then complete both trainings on topics matched for length, readability, and difficulty, followed by a short exam measuring comprehension (the primary outcome). Participants self-rate confidence and perceived cognitive load (Paas, 1992) after each format. The protocol will undergo IRB review with consent protecting disability disclosure. For RQ3–RQ4, data are collected throughout pipeline development using the error, time, and iteration metrics above, benchmarked against the conventional workflow.

Deploying this open-source pipeline is anticipated to reduce the time required to generate accessible training materials relative to conventional workflows. For learners, the audiovisual format is anticipated to yield higher comprehension, greater confidence, and lower perceived cognitive load.

This research aims to provide HRD professionals with an open-source blueprint for creating accessible training materials without adding to the manual workload of the instructional design staff. Feasibility criteria include accurate reproduction, professional quality on human review, and production effort at or below baseline. Theoretically, the study extends multimedia learning and cognitive load research to neurodiverse workplace training.

Keywords: Dyslexia accommodations, generative artificial intelligence, inclusive instructional design

References

CAST. (2024). Universal Design for Learning Guidelines version 3.0. https://udlguidelines.cast.org

Charness, G., Gneezy, U., & Kuhn, M. A. (2012). Experimental methods: Between-subject and within-subject design. *Journal of Economic Behavior & Organization, 81*(1), 1–8. https://doi.org/10.1016/j.jebo.2011.08.009

Gemma Team. (2025). *Gemma 3 technical report* (arXiv:2503.19786). arXiv. https://arxiv.org/abs/2503.19786

HaCohen, Y., Brazowski, B., Chiprut, N., Bitterman, Y., Kvochko, A., Berkowitz, A., Shalem, D., Lifschitz, D., Moshe, D., Porat, E., Richardson, E., Shiran, G., Chachy, I., Chetboun, J., Finkelson, M., Kupchick, M., Zabari, N., Guetta, N., Kotler, N., … Farbman, Z. (2026). *LTX-2: Efficient joint audio-visual foundation model* (Version 1). arXiv. https://doi.org/10.48550/ARXIV.2601.03233

Ingesson, S. G. (2007). Growing up with dyslexia: Interviews with teenagers and young adults. *School Psychology International, 28*(5), 574–591. https://doi.org/10.1177/0143034307085659

Kast, M., Meyer, M., Vögeli, C., Gross, M., & Jancke, L. (2007). Computer-based multisensory learning in children with developmental dyslexia. *Restorative Neurology and Neuroscience, 25*(3–4), 355–369. https://doi.org/10.3233/RNN-2007-253417

Kizilaslan, A., & Tunagür, M. (2021). Dyslexia and working memory: Understanding reading comprehension and high level language skills in students with dyslexia. *Kastamonu Eğitim Dergisi, 29*(5), 941–952. https://doi.org/10.24106/kefdergi.741028

Knoop-van Campen, C. A. N., Segers, E., & Verhoeven, L. (2018). The modality and redundancy effects in multimedia learning in children with dyslexia. *Dyslexia, 24*(2), 140–155. https://doi.org/10.1002/dys.1585

Le Cunff, A.-L., Martis, B.-L., Glover, C., Ahmed, E., Ford, R., Giampietro, V., & Dommett, E. J. (2025). Cognitive load and neurodiversity in online education: A preliminary framework for educational research and policy. *Frontiers in Education, 9*. https://doi.org/10.3389/feduc.2024.1437673

Mayer, R. E. (2020). *Multimedia learning* (3rd ed.). Cambridge University Press.

Paas, F. G. W. C. (1992). Training strategies for attaining transfer of problem-solving skill in statistics: A cognitive-load approach. *Journal of Educational Psychology, 84*(4), 429–434. https://doi.org/10.1037/0022-0663.84.4.429

Panjwani-Charania, S., & Zhai, X. (2024). AI for students with learning disabilities: A systematic review. In X. Zhai & J. Krajcik (Eds.), *Uses of artificial intelligence in STEM education* (1st ed., pp. 469–493). Oxford University Press. https://doi.org/10.1093/oso/9780198882077.003.0021

Schulte-Körne, G. (2010). The prevention, diagnosis, and treatment of dyslexia. *Deutsches Ärzteblatt International*, 107(41), 718–727. https://doi.org/10.3238/arztebl.2010.0718

Smythe, I., & Everatt, J. (2001). Adult check list. In I. Smythe (Ed.), *The dyslexia handbook 2001*. British Dyslexia Association.

Sweller, J. (2011). Cognitive load theory. In J. P. Mestre & B. H. Ross (Eds.), *Psychology of learning and motivation* (Vol. 55, pp. 37–76). Academic Press. https://doi.org/10.1016/B978-0-12-387691-1.00002-8

Wissell, S., Karimi, L., Serry, T., Furlong, L., & Hudson, J. (2022). “You don’t look dyslexic”: Using the job demands—resource model of burnout to explore employment experiences of Australian adults with dyslexia. *International Journal of Environmental Research and Public Health, 19*(17), 10719. https://doi.org/10.3390/ijerph191710719